

An Internship Report

Of

Artificial Intelligence

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech (CSE)



By

Batch / Roll: Section 2CSE7 | Roll No. 25SCS1003002660

Name of Student: ARYAN PRAJAPATI

Academic Year: 2025–29

CANDIDATE'S DECLARATION

I, ARYAN PRAJAPATI, do hereby declare that the internship report titled "Artificial Intelligence Internship" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AIML). All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:- Aryan

Student Name: ARYAN PRAJAPATI

Batch / Roll No.: Section 2CSE7 / 25SCS1003002660

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Codec Technologies Pvt. Ltd. and Dr. Anurag Shrivastava for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Artificial Intelligence & Machine Learning. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering, for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:

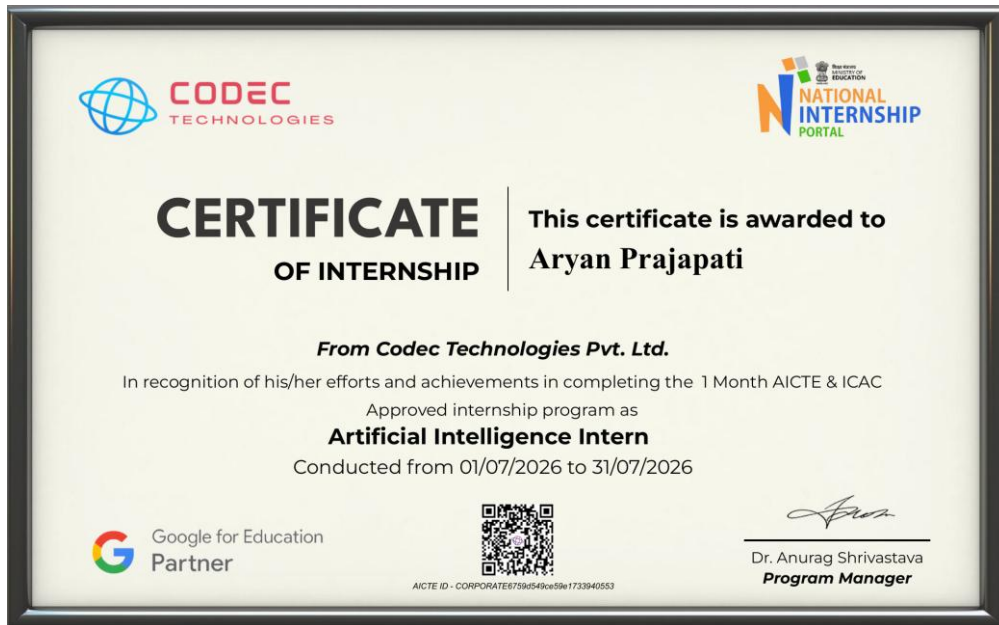
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3. INTERNSHIP COMPLETION CERTIFICATE



Internship Offer Letter



INTERNSHIP OFFER LETTER

Dear Aryan Prajapati ,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration : 01/07/2026 to 31/07/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.
Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 01/07/2026



Dr. Anurag Shrivastava
Talent Acquisition Manager



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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 1-Month Internship undertaken at Codec Technologies Pvt. Ltd., in the domain of "Artificial Intelligence". The internship was carried out from 01 July 2026 to 31 July 2026, in a hybrid, Pan-India mode, as part of the requirement of the Bachelor of Technology (Computer Science and Engineering – AIML) programme at IILM University, Greater Noida. The primary aim of this internship was to bridge the gap between classroom machine learning theory and the practical, hands-on skills required to design, build, and evaluate complete, real-world predictive models.

The internship followed a structured, week-wise curriculum that combined conceptual learning in Python, data preprocessing, and machine learning algorithms with hands-on project work, culminating in a complete, end-to-end regression project: a House Price Prediction model. On successful completion of all the requirements, an Internship Completion Certificate was awarded by Codec Technologies Pvt. Ltd. under the AICTE & ICAC approved internship program listed on the National Internship Portal.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. is an ISO 9001:2015 certified, global IT and business consultancy platform, dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth. The organization is a recognized Google for Education Partner and conducts its internship programs under the AICTE & ICAC approved framework, listed on the AICTE National Internship Portal, to help engineering students strengthen their employability and gain practical exposure to industry-relevant artificial intelligence and machine learning practices.

4.3 Problem Statement

Although a large number of engineering students learn machine learning theory and algorithms during their academic coursework, very few get hands-on exposure to building complete, real-world predictive models from scratch. Writing correct model code in isolation is only one part of applied machine learning; cleaning and exploring data, engineering meaningful features, training and comparing algorithms, and evaluating results rigorously is an equally important and often overlooked skill set. The problem addressed by this internship was to understand and practically implement the complete, end-to-end process of applied machine learning — from data cleaning and exploration to feature engineering, model training, and evaluation.

4.4 Project Objectives

- To understand the complete applied machine learning lifecycle, from data collection to model evaluation.
- To clean, explore, and preprocess real-world datasets using Python and Pandas.
- To apply feature engineering techniques, including handling missing values and encoding categorical variables.
- To gain hands-on experience with regression algorithms such as Linear Regression, Decision Tree, and Random Forest.
- To evaluate model performance using standard regression metrics (RMSE, MAE, R^2).
- To learn version control using Git and manage source code/notebooks on GitHub.
- To tune hyperparameters and improve model accuracy.
- To apply all the above concepts in a real, end-to-end project: a House Price Prediction model.

4.5 Scope of the Project

The scope of this internship spans the core applied machine learning pipeline — starting from data cleaning and exploratory data analysis, through feature engineering and model training, to evaluation and result interpretation. It covers the development of one complete regression project: a House Price Prediction model that estimates property prices from features such as area, location, bedrooms, and bathrooms. The internship was limited to model development in a hybrid, self-paced virtual/lab environment provided by Codec Technologies and did not involve production deployment, backend integration, or live data pipelines.

4.6 Technologies and Tools Used

- Languages: Python 3, Pandas, NumPy
- Editor & Workflow: Jupyter Notebook, VS Code
- Version Control: Git, GitHub
- Deployment: Streamlit / GitHub

4.7 Key Projects Undertaken

Project 1: House Price Prediction — Model Development

A supervised machine learning regression model built to predict house prices from features such as area, location, number of bedrooms and bathrooms, using a cleaned real-estate dataset.

Key features include exploratory data analysis to identify key price-driving features, data cleaning with handling of missing values and outlier treatment, feature engineering and encoding of categorical variables, and regression model training using Linear Regression and Random Forest. Skills demonstrated: data preprocessing, feature engineering, regression modeling, and model evaluation.

Project 2: House Price Prediction — Evaluation & Results

A structured ML pipeline covering data ingestion, preprocessing, model training, and evaluation, benchmarked using standard regression metrics to validate prediction accuracy on unseen housing data. Key features include a train-test split with cross-validation for reliable evaluation, model comparison across Linear Regression, Decision Tree, and Random Forest, performance evaluated using RMSE, MAE, and R^2 score, and a final model saved and tested on sample house price predictions. Skills demonstrated: model evaluation, hyperparameter tuning, Scikit-learn, and data visualization.

4.8 Methodology / Weekly Timeline

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on practice, and project submissions, as summarized in the table below.

Week	Module	Description
1	Onboarding & Fundamentals	Program onboarding, Python & ML fundamentals refresher, and environment setup with Jupyter Notebook and Git.
2	Data Preprocessing & EDA	Data cleaning, exploratory data analysis, and feature engineering on the housing dataset.
3	Project Build — Model Training	Trained and compared regression models for the House Price Prediction project.
4	Evaluation & Wrap-up	Evaluated model performance, tuned hyperparameters, and completed final review.

Each week combined guided learning material with practical, hands-on project work. In the final week, the model was evaluated, tested, and documented, and an overall internship review was conducted by the assigned project head.

4.9 Expected Outcomes

- Ability to independently clean, explore, and preprocess real-world datasets using Python and Pandas.
- Practical understanding of regression algorithms and their comparative performance.

- Hands-on exposure to feature engineering and categorical variable encoding.
- Capability to manage source code and notebooks using Git and GitHub.
- Skill to evaluate and tune machine learning models using standard regression metrics.
- One completed, end-to-end machine learning project: a House Price Prediction model.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by Codec Technologies Pvt. Ltd., confirming successful completion of the 1-Month AICTE & ICAC Approved Internship Program in Artificial Intelligence (Artificial Intelligence Intern), are attached in Chapter 3 of this report.

5. BIBLIOGRAPHY/REFERENCES

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